

LAPORAN PRAKTIKUM
MODUL 4: MEMBACA
SOURCE CODE XINU



Disusun Oleh:

Dhiemas Tulus Ikhsan

2311104046 S1SE-07-02

Dosen Pengampu:

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK
FAKULTAS INFORMATIKA
TELKOM UNIVERSITY PURWOKERTO
2026

JURNAL

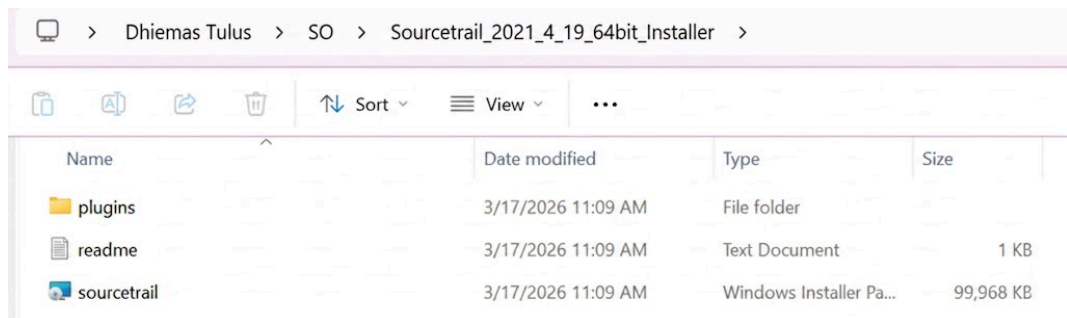
1. [10 Poin] Apa nama image yang dihasilkan setelah melakukan kompilasi pada Xinu? Berapa ukuran file tersebut? Ada pada folder apa file image tersebut?

Hint: baca kembali modul-modul sebelumnya

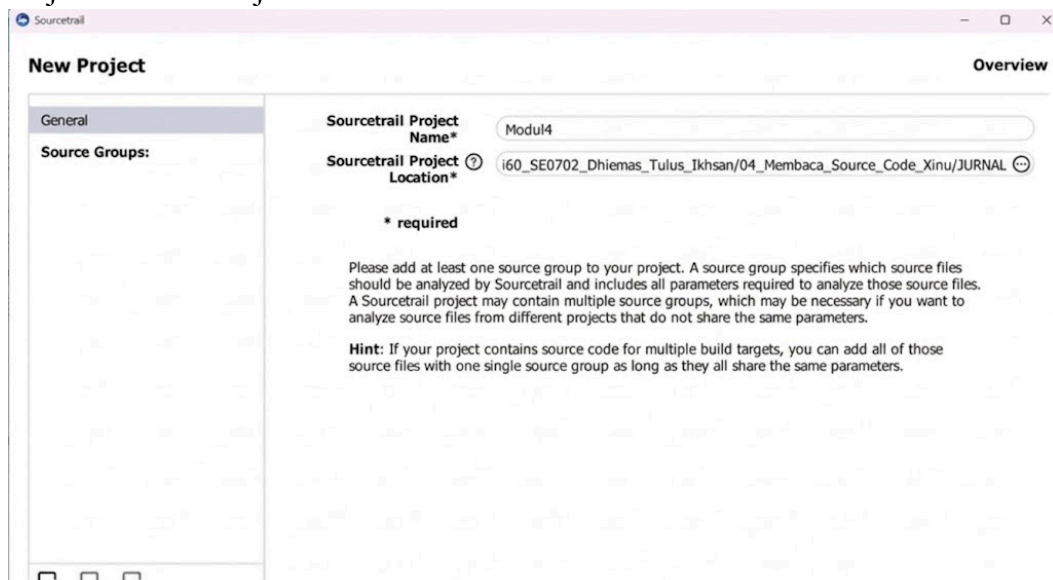
Jawaban: Nama image: biasanya bernama xinu.elf atau xinu.bin (tergantung konfigurasi build). Lokasi file: berada di folder: xinu/compile/. Ukuran file: bervariasi ($\pm$ ratusan KB, tergantung build & konfigurasi)

2. Membaca source code Xinu

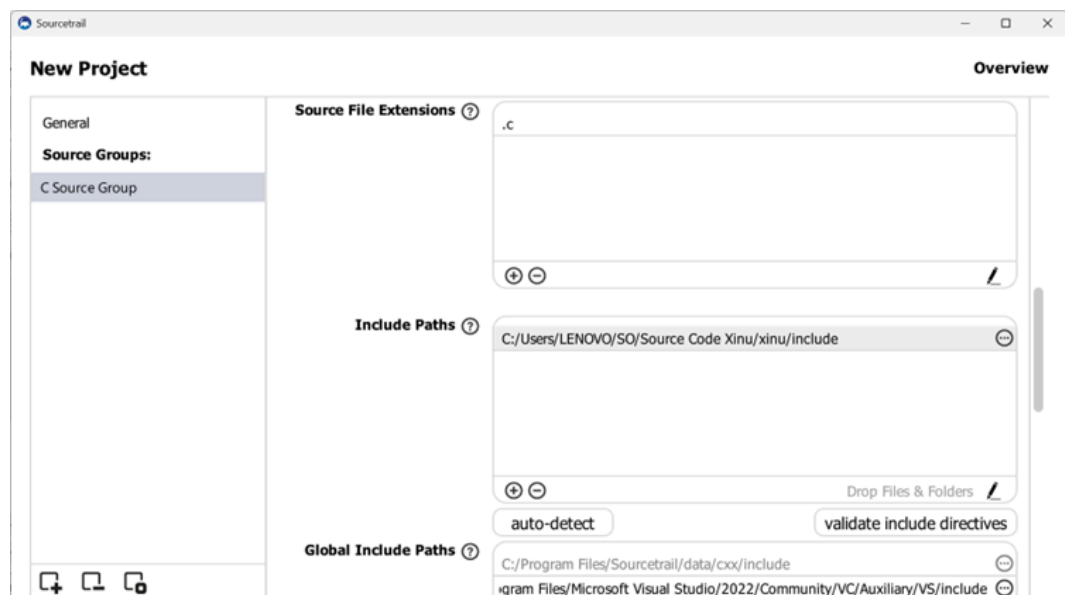
- a. Cek aplikasi bernama Sourcetrail di PC. Jika belum ada, download SourceTrail pada (DOWNLOAD SOURCE TRAIL). SourceTrail adalah software untuk mengeksplorasi source code. Programmer yang bagus lebih banyak membaca kode daripada menulis kode



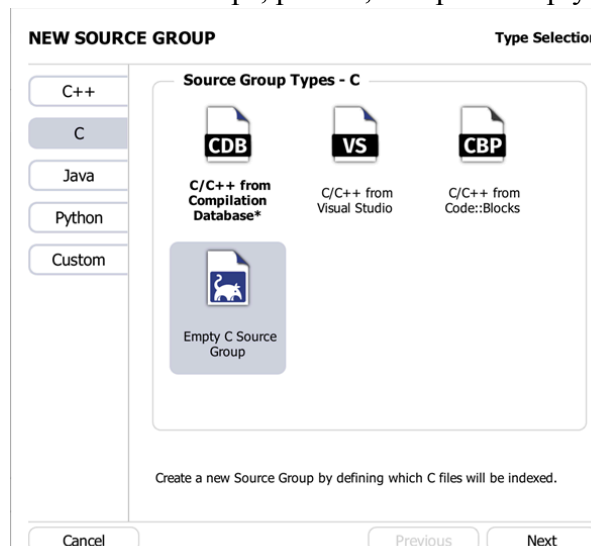
- b. Jalankan SourceTrail
- c. Project → New Project



- d. Isi nama project xinu dan pilih lokasi project di manapun



- e. Add Source Groups, pilih C, lalu pilih Empty C Source Group



- The screenshot displays the Sourcetrail IDE interface across three different file views. Each view consists of a project graph on the left, a code editor in the center, and an error log at the bottom.

 - Overview View:** The project graph shows nodes for Structs (85), Functions (879), Global Variables (140), Unions (1), and Typedefs (43). The code editor shows the 'Modul4' directory with statistics: last indexed: 2026-04-06 22:21:11, 282 files (271 complete), 26467 lines of code, 3389 symbols, 9708 references, and 53 errors (1 fatal). The error log shows one error: 'ERROR unknown type name "int32"' at line 6, column 1 in file 'C:/Users/LENOVO/SO/Source Code Xinu/xinu/include/stdio.h'.
 - rdsprocess.c View:** The project graph shows a detailed view of the 'rdsprocess.c' file and its references. The code editor shows the first few lines of the file, including a comment and an include statement. The error log shows the same error: 'ERROR unknown type name "int32"' at line 6, column 1.
 - create.c View:** The project graph shows a detailed view of the 'create.c' file and its references. The code editor shows the first few lines of the file, including a comment and an include statement. The error log shows the same error: 'ERROR unknown type name "int32"' at line 6, column 1.

The error log at the bottom of each view is structured as follows:

Type	Message	File	Line	Column
ERROR	unknown type name "int32"	C:/Users/LENOVO/SO/Source Code Xinu/xinu/include/stdio.h	6	1

```
xinu@xinu-develop-end:~$ grep -r "percent\|prstate\|prprio\|prstk" /home/xinu/xinu/include/process.h
/* Must be odd to make PR_FREE)
(struct tbl.x).prstate == PR_FREE)
#define NDESC 5 /* Must be odd to make PR_FREE)
struct percent { /* Entry in the process table */
    uint16 prstate; /* Process state: PR_CURR, etc. */
    pri16 prprio; /* Process priority */
    char *prstkptr; /* Saved stack pointer */
    char *prstbase; /* Base of run time stack */
    uint32 prstlen; /* Stack length in bytes */
extern struct percent proctab[];
xinu@xinu-develop-end:~$ grep -r "struct percent" /home/xinu/xinu/include/process.h
struct percent { /* Entry in the process table */
xinu@xinu-develop-end:~$ cat /home/xinu/xinu/include/process.h
/* process.h - isbadpid */

/* Maximum number of processes in the system */

#ifndef NPROC
#define NPROC 8
#endif

/* Process state constants */

#define PR_FREE 0 /* Process table entry is unused */
```

- [illegible]

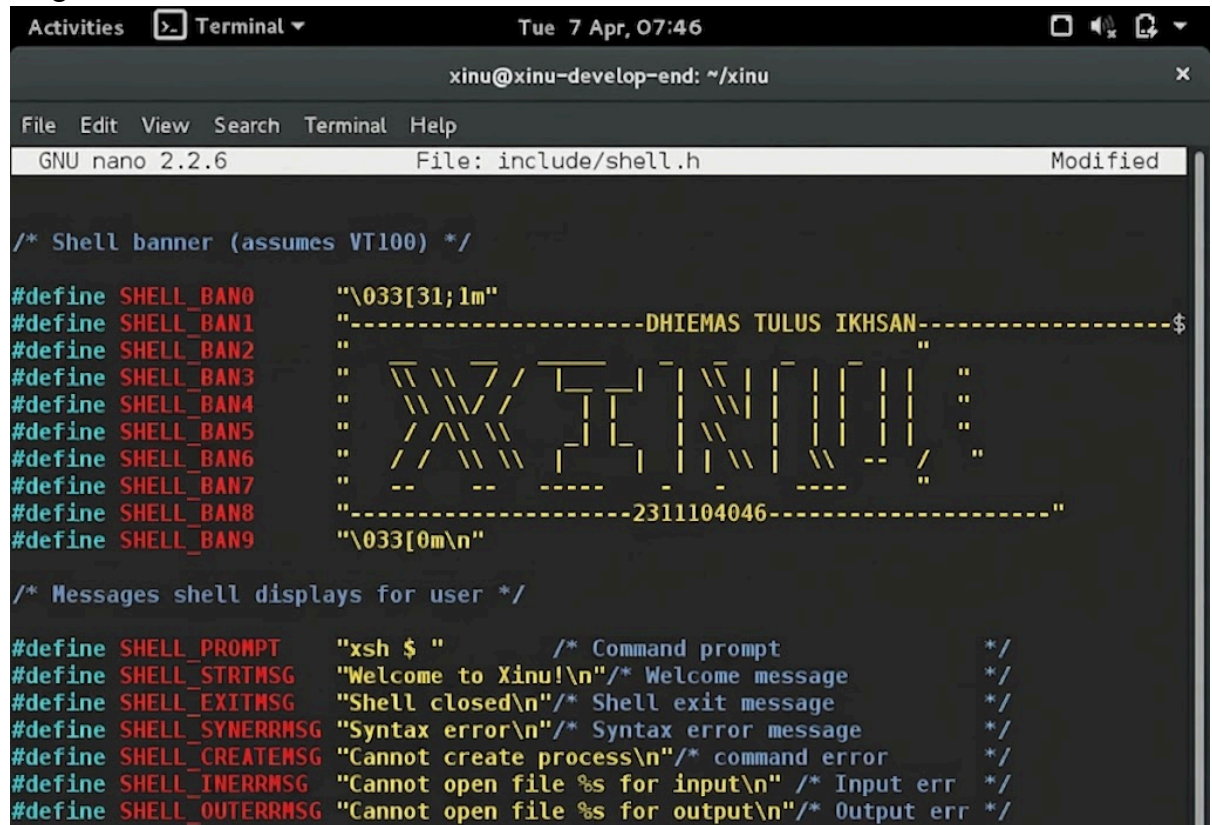
- ```
xinu@xinu-develop-end: "
File Edit View Search Terminal Help
xinu@xinu-develop-end:~$ clear
[3;J
xinu@xinu-develop-end:~$ grep -rl "banner\\Welcome\\XINU" /home/xinu/xinu/shell/ --include="*.c"
/home/xinu/xinu/shell/shell.c
xinu@xinu-develop-end:~$ cat /home/xinu/xinu/shell/shell.c
/* shell.c - shell */

#include <xinu.h>
#include <stdio.h>
#include "shprototypes.h"

/*
 * Table of Xinu shell commands and the function associated with each
 */
const struct cmdent cmdtab[] = {
 {"argecho", xsh_argecho},
 {"arp", xsh_arp},
 {"cat", xsh_cat},
 {"clear", xsh_clear},
 {"date", xsh_date},
 {"devdump", xsh_devdump},
 {"echo", xsh_echo},
 {"help", xsh_help},
 {"ls", xsh_ls},
 {"kill", xsh_kill},
 {"memdump", xsh_memdump},
 {"memstat", xsh_memstat},
 {"ns", xsh_ns},
 {"netinfo", xsh_netinfo},
 {"ping", xsh_ping},
 {"ps", xsh_ps},
 {"sleep", xsh_sleep},
 {"tee", xsh_tee},
 {"udp", xsh_udpdump},
 {"udpecho", xsh_udpecho},
 {"udpserver", xsh_udpserver},

```

- c. Ubahlah welcome banner Xinu sehingga menjadi seperti ini:  
 Tambahkan NAMA dan NIM masing-masing, kemudian ubah banner Xinu sesuai dengan selera Anda



```

/* Shell banner (assumes VT100) */

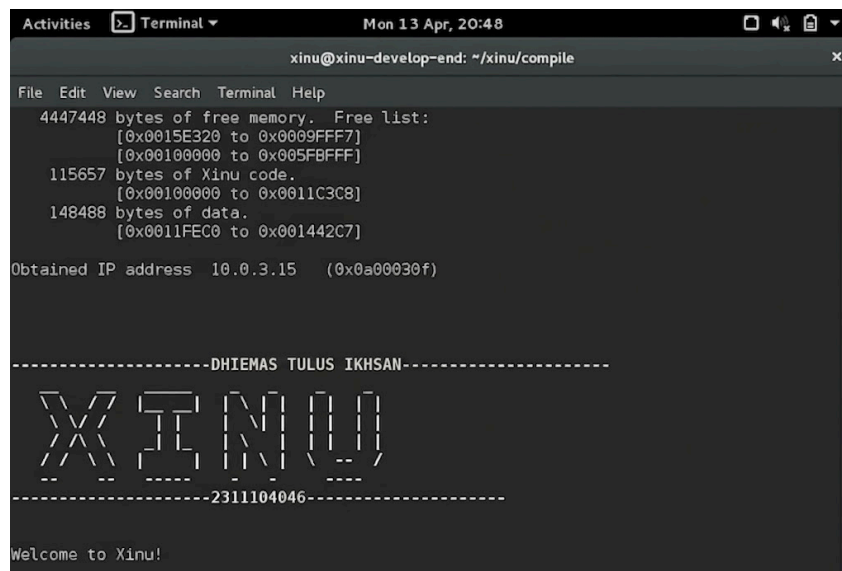
#define SHELL_BAN0 "\033[31;1m"
#define SHELL_BAN1 "-----DHIEMAS TULUS IKHSAN-----"
#define SHELL_BAN2 " "
#define SHELL_BAN3 " X \ \ / \ / _ _ | _ \ \ / \ / _ _ | _ \ \"
#define SHELL_BAN4 " \ \ / \ / \ \ / \ | | | | | | | | | | | | | | | \"
#define SHELL_BAN5 " / \ \ / \ / \ \ / \ | | | | | | | | | | | | | | | \"
#define SHELL_BAN6 " _ _ | _ \ \ / \ / _ _ | _ \ \ / \ / _ _ | _ \"
#define SHELL_BAN7 " _ _ | _ \ \ / \ / _ _ | _ \ \ / \ / _ _ | _ \"
#define SHELL_BAN8 "-----2311104046-----"
#define SHELL_BAN9 "\033[0m\n"

/* Messages shell displays for user */

#define SHELL_PROMPT "xsh $ " /* Command prompt */
#define SHELL_STRMSG "Welcome to Xinu!\n" /* Welcome message */
#define SHELL_EXITMSG "Shell closed\n" /* Shell exit message */
#define SHELL_SYNERRMSG "Syntax error\n" /* Syntax error message */
#define SHELL_CREATMSG "Cannot create process\n" /* command error */
#define SHELL_INERRMSG "Cannot open file %s for input\n" /* Input err */
#define SHELL_OUTERRMSG "Cannot open file %s for output\n" /* Output err */

```

- d. Setelah melakukan perubahan lakukan perintah berikut ini pada terminal Linux
- \$ cd xinu/compile
  - \$ make clean
  - \$ make
  - \$ sudo minicom
  - Jalankan vm backend



```

4447448 bytes of free memory. Free list:
[0x0015E320 to 0x0009FFFF]
[0x00100000 to 0x005FBFFF]

115657 bytes of Xinu code.
[0x00100000 to 0x0011C3C8]

148488 bytes of data.
[0x0011FEC0 to 0x001442C7]

Obtained IP address 10.0.3.15 (0xa00030f)

-----DHIEMAS TULUS IKHSAN-----

 X \ \ / \ / _ _ | _ \ \ / \ / _ _ | _ \
 \ \ / \ / \ \ / \ | | | | | | | | | | | | | | |
 / \ \ / \ / \ \ / \ | | | | | | | | | | | | | | |
 _ _ | _ \ \ / \ / _ _ | _ \ \ / \ / _ _ | _ \
 _ _ | _ \ \ / \ / _ _ | _ \ \ / \ / _ _ | _ \

-----2311104046-----

Welcome to Xinu!

```